

1. Discuss algorithm and complexity

Algorithm is a detailed step-by-step process of problem. In the context of programming it is the instruction run by the computer to do an action or solve a problem such as check if age is greater than 12 and do the following instructions if true.

Complexity details the time it will take to execute the program and how much space it will consume. The standard is using the big O notation. Example: Time: $O(n)$ and Space: $O(n!)$

2. Differentiate Algorithm and Programming

Algorithm is a detailed step-by-step process of problem.

Programming is using a language specific to computers in order to solve a problem or do an action.

The key difference is algorithm is not limited in the scope of computing which means it can be applied to solve other problems such as cooking, and in the field of programming means the steps the computer uses to solve a specific problem while programming is the act of writing instructions for the computer.

3. Give at least 3 real life application of algorithm

a.) Creating a seating arrangement system to create an alphabetical seating arrangement of students.

b.) Baking one's favorite cake because the steps to bake are necessary.

c.) Preparing the things needed to go to school.

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